Ernest Ma

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Education

San Jose State University **December 2026**

Bachelor of Science in Computer Engineering | GPA: 3.95 *San Jose, CA*

- **Relevant Coursework**: Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks

Experience

KLA | Software Engineer Intern Ann Arbor, MI | May 2026 – August 2026

- Accelerated physics-ML training via **PyTorch DDP** + **ClearML** on **Kubernetes**, scaling 3.9× throughput on 4 GPUs (98% efficiency)
- Deployed a **Kubernetes** AI agent **(kagent + MCP + Prometheus)** with 12 custom tools for health scans, alert triage, and reports
- Built a **React/FastAPI** dashboard with **PostgreSQL** and **Prometheus** for **K8s** clusters, **ClearML** experiments, and **VMware**
- Standardized a **LangGraph** template and Kubernetes deployment workflows, enabling build-to-deploy in a 2-hour test

KLA | Software Engineer Intern Ann Arbor, MI | May 2025 – August 2025

- Built a log-based troubleshooting agent using **LangGraph**, **FastAPI**, and **Gemma 3 / GPT-4o** to assist field engineers
- Replaced hours of manual historical-log searching with sub-second retrieval using a **ChromaDB** RAG pipeline
- Deployed and configured the troubleshooting application on **AWS EC2 (Linux)** for **99.9%** uptime field access

SJSU College of Engineering | Software Engineer Intern San Jose, CA | January 2026 – May 2026

- Architected a **Go/Gin** event service for 1k+ users with **MongoDB** for event registrations, waitlists, and registration forms
- Mitigated data races during **10k+ write/sec** spikes by implementing **Redis** concurrency locks and optimized caching strategies
- Orchestrated a **Kafka** event-driven pipeline to decouple API requests from background registration processing

Spartan Racing: Formula SAE | Software Engineer Intern San Jose, CA | January 2026 – May 2026

- Expanded a **C-based** hardware-in-the-loop injection system for testing using CAN protocols for real-time VCU validation
- Developed a **Python** HIL test suite utilizing state machines to manage complex CAN communication sessions
- Synchronized automated telemetry validation to ensure data consistency during high-speed testing cycles
- Verified Launch Control accuracy by hand-calculating torque outputs against parameterized test cases

Projects

Ticker Trace — *Python, FastAPI, Redis Streams, PostgreSQL, Next.js, Docker* **</> Code**

- Built an **asyncio** stock-execution simulator streaming live Alpaca data through distributed **Redis Streams** processing workers
- Processed **120K** generated events at **200/sec** with **8.06 ms p95** persistence latency and zero lost or duplicate market events
- Built a **Next.js** replay lab with immutable recordings, quote-linked fills, and first-divergence latency comparisons

Agent Faultline — *Rust, Python, MCP, PostgreSQL, Docker, OpenTelemetry* **</> Code**

- Built **Rust** / **Python MCP** benchmarks with 5 fault types across **PostgreSQL**-backed payment and deployment environments
- Engineered a **Rust** proxy and scoring pipeline; verified idempotency and revision guards in 10K-attempt tests
- Added scoped **stdio** / **HTTP MCP** support with capability checks, fault injection, and trace-based recovery reports

PatchBench — *Python, pytest, Docker, YAML* **</> Code**

- Built an executable benchmark to measure whether AI code reviewers detect regressions that pass visible tests
- Designed an 18-task **Python** corpus with 36 correct and regression patches, test isolation, and **Docker-based** replay
- Implemented a CLI reviewer harness for structured findings and deterministic detection, false-positive, and inconclusive scoring

Collab Code — *TypeScript, VS Code Extension API, Liveblocks, Yjs* **</> Code**

- Built a real-time collaborative IDE extension using Yjs CRDTs and Liveblocks for multi-user editing and room sessions
- Implemented synchronized cursors, selections, and presence indicators with document binding and local-origin filtering
- Created account-less shared-key onboarding and automated VSIX packaging for seamless cross-editor distribution

Technical Skills

Languages: Python, Go, TypeScript, Rust, SQL, C++, C, Java
Frameworks & Libraries: FastAPI, React, Gin, PyTorch, LangGraph, Yjs, Next.js
Infrastructure & Cloud: Kubernetes, Docker, AWS EC2, Kafka, Redis Streams, Prometheus, OpenTelemetry, Helm, Linux
Databases: PostgreSQL, MongoDB, Redis, SQLite, ChromaDB